SHARP

PERSONAL COMPUTER

MZ - 80B

DISK BASIC MANUAL

SB-6511

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NOTICE

This manual is applicable to the SB-6511 DISK BASIC interpreter used with the SHARP MZ-80B Personal Computer. The MZ-80B general-purpose personal computer is supported by system software which is filed in software packs (cassette tapes or diskettes).

All system software is subject to revision without prior notice, therefore, you are requested to pay special attention to file version numbers.

This manual has been carefully prepared and checked for completeness, accuracy and clarity. However, in the event that you should notice any errors or ambiguities, please feel free to contact your local Sharp representative for clarification.

All system software packs provided for the MZ-80B are original products, and all rights are reserved. No portion of any system software pack may be copied without approval of the Sharp Corporation.

Introduction

Before handling floppy diskettes, please read carefully the notes in the section, "Handling Diskettes" on page 37.

The master diskette and blank diskette will not be exchanged for new ones after purchase.

It is recommended that the master diskette be copied using the disk copy utility (refer to another mannual) to generate a submaster diskette, and that the submaster diskette be used for day to day operation. Be sure to keep the master diskette in a safe place.

Master Diskette Protection

A write protect seal is affixed to the write protect notch of the master diskette to prevent its contents from being accidentally erased through erroneous operation or accidents such as a power failure.

Never remove the write protect seal; if it is damaged, replace it with a new one.

Contact your dealer for assistance if you should find any ambiguities in this manual.

Change of diskettes under operation

While the computer is accessing the disk for reading/writing files, never change with another diskette before executing KILL command.

If changed in such a state, the contents of the diskette will be destroyed.

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General Information

BASIC SB-6511 is an improved version of Disk BASIC that is based entirely on DISK BASIC SB-6510 and includes control statements related to the GP-IB and RS-232C interfaces. This BASIC can be used to control equipment with the MZ-80B computer through the GP-IB interface card MZ-8BI04 and serial interface card MZ-8BI03.

For this version of BASIC there is another instruction manual in addition to this one. This manual deals only with those portions of the BASIC language which are concerned with the GP-IB and RS-232C interfaces. There is another manual concerning the SB-6510 version of the BASIC language which deals with the BASIC language in general. SB-6511 BASIC can also be used in the same way. Reference a separate manual for information concerning how to activate BASIC, copy disks and similar operations.

In this manual the first chapter deals with GP-IB BASIC statements, the second chapter with RS-232C statements, and the Appendix contains an ASCII code list, error message list, and information concerning how to handle diskettes.

Chapter 1
GP-IB Statements

This chapter contains information on the BASIC used to control measuring equipment with the GP-IB interface card MZ-8B104.

Reference the GP-IB interface manual for details concerning the GP-IB Interface.

Statement format

Statements must be coded according to the following conventions.

- *Small letters and reverse characters cannot be used for any commands and statements.*
- *Operands which must be specified by the programmer are indicated in italics.*
- *Items in brackets "( )" may be omitted or repeated any number of times. However, the bracket marks should not be typed when marking the relevant input.*
- *Separators (commas, semicolons, etc.) must be correctly placed in the specified positions.*

With BASIC language, programs can be made without paying special attention to the commands peculiar to most GP-IB. However, commands based on the standard have predetermined order of transmission, and once an error is made in this order, instruments will not be operated.

Besides, all the instruments have their respective addresses. If more than one instruments having the same address exist in one system, wrong operation may be caused.

Furthermore, controlled instruments have their own codes corresponding to their operations. Different codes are used for different instruments, except for same instruments manufactured by the same company. Consequently, it is necessary to read carefully the Operation Manual of each device before starting programming. The same applies to codes which indicate end of data. End of data can be indicated by EOI, ASCII code (instead of EOI), or the combination of both. All these details are peculiar to each instrument, and cannot be covered by the standard. They are not included in BASIC commands either, but are contained in string data to be transmitted as "data". All these must be taken into consideration in programming.

List of GP-IB Statements

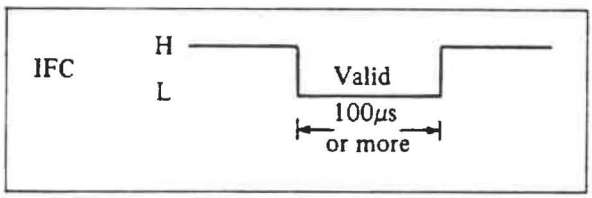
Statement	Description
1. ICL	MZ-80B as system controller initializes the whole interface.
2. REN	MZ-80B as system controller sets the whole interface in remote mode.
3. LCL	MZ-80B as system controller sets the whole interface in local mode.
4. LCL n	MZ-80B as controller resets the instrument designated by n on the interface bus in local mode.
5. LLO	MZ-80B as controller prohibits all the instruments on the interface bus to be reset in local mode.
6. DCL	MZ-80B as controller sets all the instruments on the interface bus in the initial state predetermined on the equipment side.
7. DCL n	MZ-80B as controller sets the instrument designated by n on the interface bus in the initial state predetermined on the instrument side.
8. TRG n	MZ-80B as controller causes the equipment designated by n on the interface bus to start operation.
9. PCT n	MZ-80B as controller passes the right to control to the other instrument equipped with controller function on the bus. Thereafter, the operation of MZ-80B as controller constitutes an error.
10. WRT n, V	MZ-80B as controller causes the instrument designated by n to become listener and causes itself to become talker. Then, it transmits data designated by V.
11. RED n, V	MZ-80B as controller causes the instrument designated by n to become talker and causes itself to become listener. Then, it accommodates received data in V.
12. WRT/V	MZ-80B as talker transmits data designated by V.
13. RED/V	MZ-80B as listener accommodates received data in the variable designated by V.
14. CMDW AS	MZ-80B as controller issues the command designated by AS, and thereafter becomes a talker.
15. CMDR AS	MZ-80B as controller issues the command designated by AS, and thereafter becomes a listener.
16. ON SRQ	MZ-80B jumps to the processing routine when any one of the instruments on the bus is making a service request.
17. SPOL n, A	MZ-80B as controller executes serial poll for instrument n on the bus, and accommodates status byte in variable A.
18. PPC n, l	When executing parallel poll, MZ-80B designates the status bit to which the equipment on the bus responds with 1 and the instrument address with n. (Decimal designation)
19. PPOL A	MZ-80B as controller executes parallel poll, and accommodates status bit in variable A.
20. PPU	MZ-80B clears parallel poll mode.
21. GPIBM n	MZ-80B always functioning as system controller and active controller designates its own address with n.
22. EOIW n	Designates delimiter when transmitting data. Delimiters exceeding 2 are ignored. If there is only one, "," is not necessary.
23. EOIR n	Designates delimiter when receiving data. Delimiters exceeding 2 are ignored. If there is only one, "," is not necessary.

When the BASIC is actuated, system controller, active controller and Address No. 1 are set, although no I/O operation is carried out.

1. ICL *(Interface Clear)*

Format ICL

Function Negative pulse of 100μs or more is generated on the IFC line, to initialize the interface as a whole.

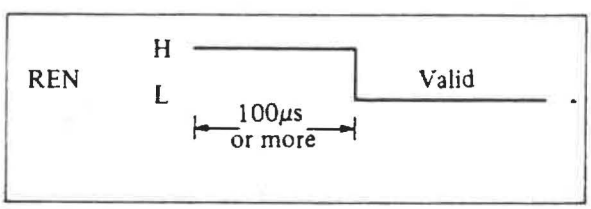


Description
- When MZ-80B is not a system controller, an error will be caused.
- Variables are not required.

2. REN *(Remote Enable)*

Format REN

Function By setting the REN line at low level after keeping it at high level for 100μs, the instrument on the bus can be set in remote mode.

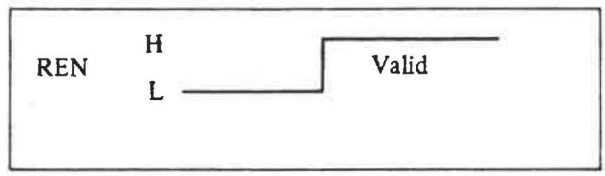


Description
- When MZ-80B is not a system controller, an error will be caused.
- Variables are not required.

3. LCL *(Local)*

Format LCL

Function The REN line is set at high level to reset all the instruments on the bus in local mode (manual mode).



Description

- When MZ-80B is not a system controller, an error will be caused.
- When controlling the instruments again, REN command must be executed.
- Variables are not required.

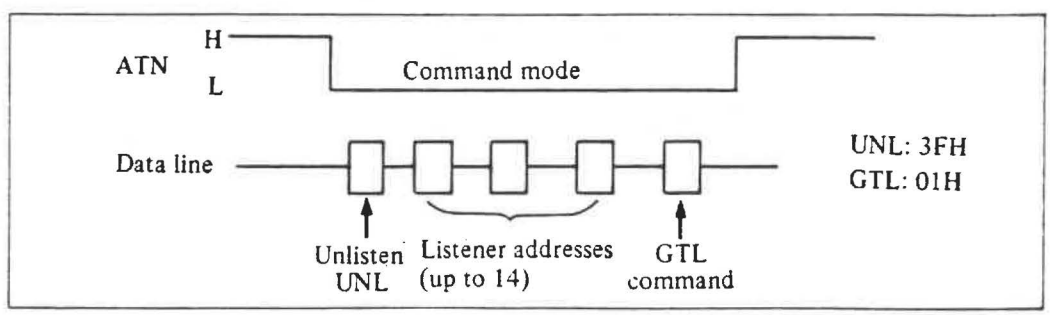
4. LCL n *(Go to local No. n)*

Format LCL $n_1<;n_2;\ldots;n_m>$

n_k: *device number (listener)*

Function

- MZ-80B sets the ATN line at low level (command mode), clears listener, issues listener address for instrument designated by n_k, transmits GTL command, and sets the designated listener in local mode.
- The designated number of listeners are set in local mode. However, m shall be 14 or less.



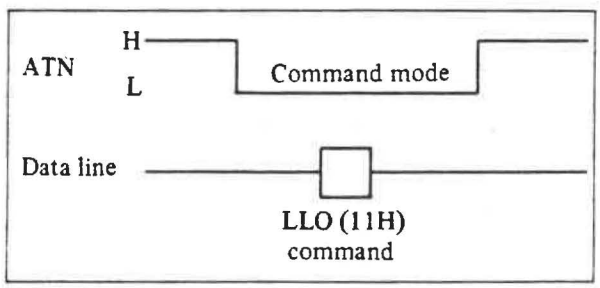
Description

- n_k shall be expressed in a decimal number from 0 to 30.
- When MZ-80B is not an active controller, an error will be caused.

5. LLO *(Local Lock-out)*

Format LLO

Function MZ-80B sets the bus in command mode and issues LLO command to prohibit all the instruments on the bus to return to local mode. (That is, the front panel key operation becomes invalid.)

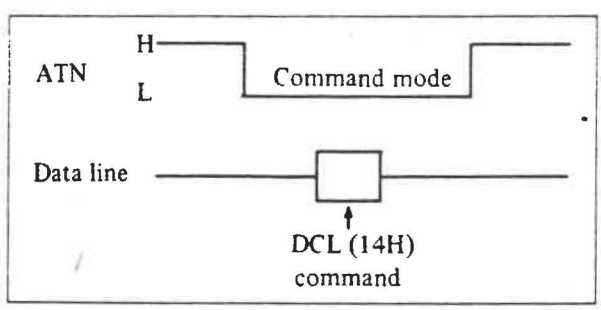


Description
- When MZ-80B is not an active controller, an error will be caused.
- Variables are not required.

6. DCL *(Device Clear)*

Format DCL

Function MZ-80B sets the bus in command mode, and issues DCL command to set all the instruments on the bus in their respective initial states.



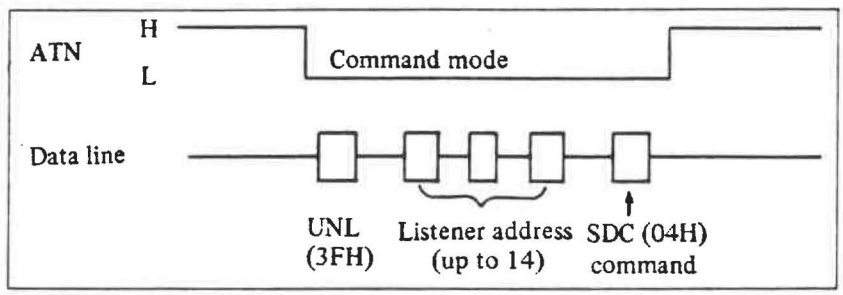
Description
- When MZ-80B is not an active controller, an error will be caused.
- Variables are not required.
- For initial states peculiar to instruments, refer to Operation Manuals of the instruments.

7. DCL n *(Selected Device Clear)*

Format DCL $n_1 <; n_2 ; \ldots ; n_m >$

n_k : *device number (listener)*

Function MZ-80B sets the bus in command mode, clears all the listeners using UNL code, sends the listener address for instrument designated by n_k, issues SDC command, and sets the designated listener in the initial state peculiar to the instrument.
The designated number of listeners can be initialized.



Description

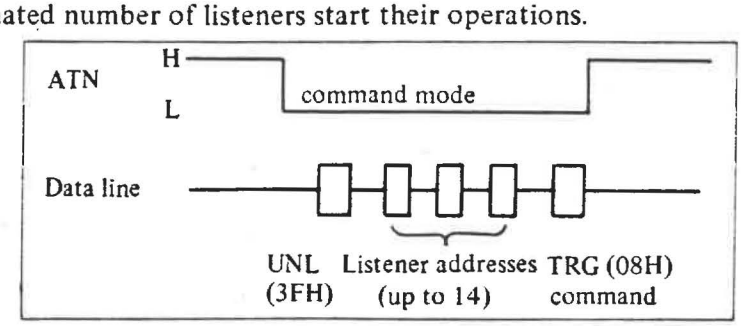
- m shall be 14 or less.
- n_k shall be expressed in a decimal number from 0 to 30.
- When MZ-80B is not an active controller, an error will be caused.

8. TRG n *(Trigger)*

Format TRG $n_1 <; n_2 ; \ldots ; n_m >$

n_k : *device number (listener)*

Function After setting the bus in command mode and clearing listeners, MZ-80B transmits listener address for instrument designated by n_k followed by TRG command to start the operation (measurement) of the designated listener.
The designated number of listeners start their operations.



Description

- m shall be 14 or less.
- n_k shall be expressed in a decimal number from 0 to 30.
- When MZ-80B is not an active controller, an error will be caused.

9. PCT n *(Pass Control)*

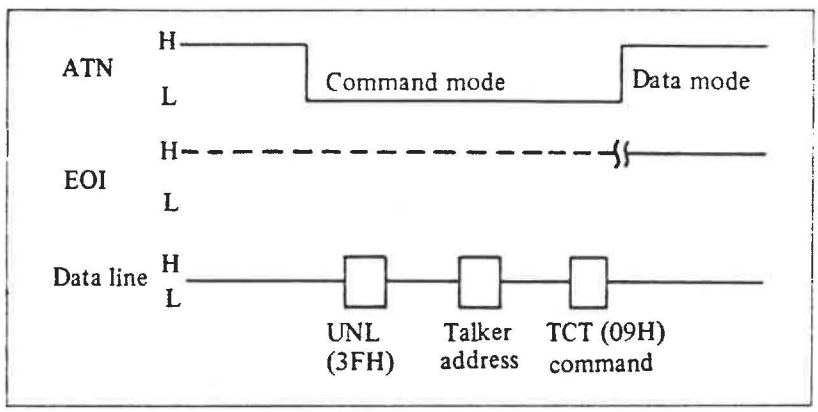
Format

PCT *n*

n: device number (talker)

Function

After setting the bus in command mode and clearing listeners, MZ-80B transmits the talker address of the instrument with controller function followed by TCT command to pass the right of controller to that instrument.



Description

- The number of devices shall be 1. The instrument with controller function shall be specified as device number.
- MZ-80B is an active controller in command mode. However, in data mode, command cannot be issued until it is designated as a mere listener or talker on the bus. It resumes the function of active controller when it executes ICL.

10. WRT n, V *(Write)*

Format

WRT n_1<, n_2;. . . . ; n_k>, V_1<, V_2, . . . , V_m>

n_i: *device number (listener)*

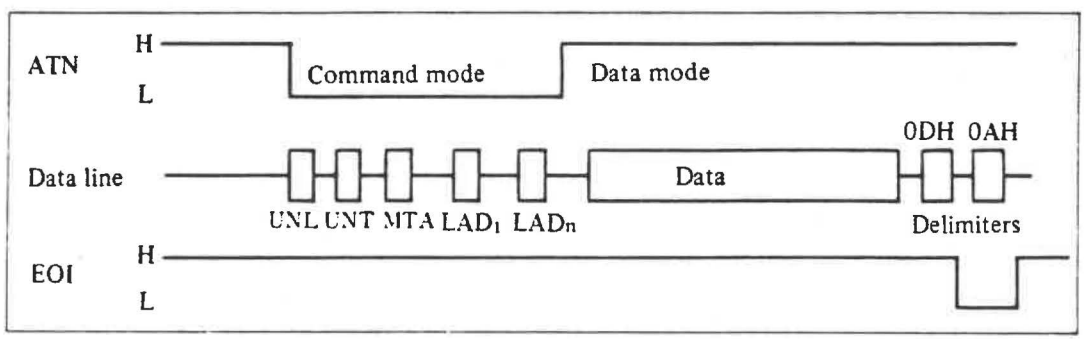
V_j: *transmit data (string variable, numeric variable or constant)*

Function

With this statement MZ-80B transmits designated data to designated listeners. MZ-80B, as a controller, designates itself as a talker in command mode and an individual device indicated by n_i as a listener.

After issuing a command, MZ-80B sets the bus in data mode and sends each designated listener data designated by V_j.

After transmitting data, MZ-80B automatically sends a delimiter code 0DH-0AH and EOI with the timing shown below.



Description

- If transmit data is composed of numerical values, it is converted into a string and output as a ASCII code. If two or more sets of transmit data are designated, they are successively output as a series of ASCII codes.
- Decimal numbers from 0 to 30 are available as device numbers.
- MZ-80B, as a talker, can designate a maximum of 14 listeners.
- This is executable only in a controller that can get active. (Controller in charge) If this is executed after pass control, an error occurs.

Example

```
10 WRT 5, "SP1;" . . . . . ."SP1;" is sent to device No. 5.

10 MZ = 3 : PC = 4 : DS = "ABC"
20 WRT MZ ; PC, DS . . . . . ."ABC" is sent to devices Nos. 3 and 4.

10 PL = 5 : PN = 4
20 WRT PL, "SP", PN, ";" . . . . . ."SP4;" is sent to device No. 5.
```

11. RED n, V *(Read)*

Format

RED *n*, V_1<, V_2, . . . , V_k>

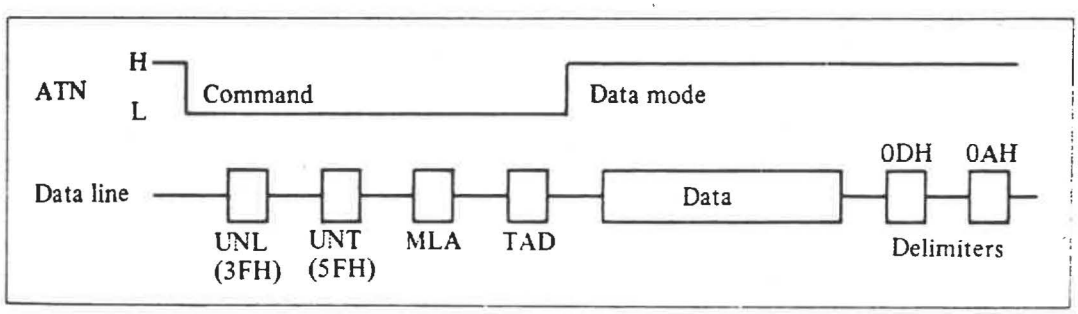
n: device number (talker)

V_i*: Variable for storing received data*

Function

With this statement MZ-80B receives data from a designated talker and then stores it in a designated variable.

In command mode MZ-80B designates itself as a listener and an individual device indicated by *n* as a talker; and in data mode it receives data sent from designated talker. MZ-80B checks the EOI line every byte. If the EOI is low in level, it receives the data present on the then data line and stops receiving any more data, while the EOI is high in level, MZ-80B stops receiving data the moment it receives a delimiter code 0DH-0AH.



Description

- MZ-80B can use only a single string variable when it is used as store variable. This is because the string variable can contain all data – a maximum of 253 characters – ahead of a delimiter.
- If numeric variables are used, MZ-80B converts all data ahead of a comma (,) or delimiter into numerical values and sequentially stores them in the corresponding variables. (If the above-mentioned conversion cannot be achieved, an error occurs.)
- When two or more numeric variables are used, the number of data must coincide with that of variables.
- The format of data sent from measuring instruments is not unified. Therefore, when receiving data by the aid of this statement, it is necessary to note the format of received data.

Example

```
10 RED 5, AS ..... Date from device No. 5 is received and
                   stored in AS.
10 FC = 24
20 RED FC, F1, F2 ..... If transmitted data is 100 and 240,
                        F1= 100 and F2= 240.
```

12. WRT/ V

Format

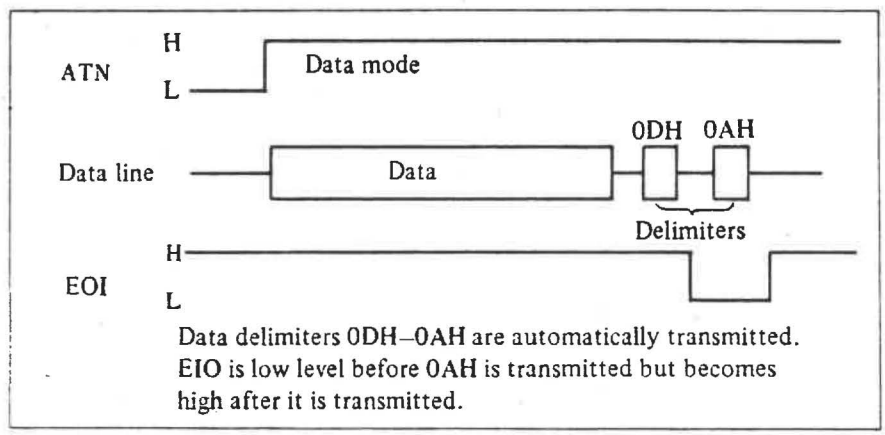
WRT/ V_1<, V_2, . . . , V_k>

V_i: *Transmit data*

Function

MZ-80B checks whether it has been designated as a talker and, if designated, transmits data. If it is not designated as a talker, an error occurs.

The format of transmit data is the same as that of statement "WRT n, V." When transmitting much data to the same device, it is practical to designate some devices as a talker or listener. This will enable to perform data transmission only in data mode with statement "WRT/V" because these devices can serve as a talker or listener until they are cancelled or re-designated.



Data delimiters 0DH–0AH are automatically transmitted.
EIO is low level before 0AH is transmitted but becomes high after it is transmitted.

Example

```
10 WRT 9, AS
20 FOR I = 1 TO 100
30 WRT/AS
40 NEXT I
```

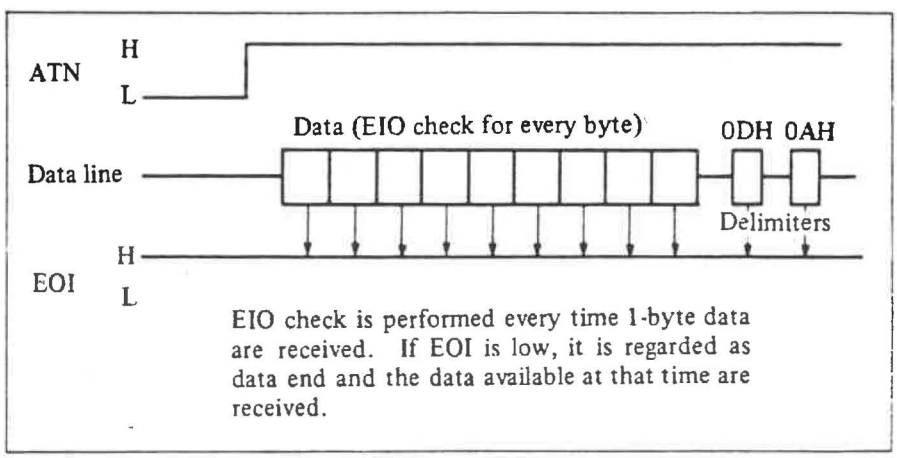
The contents of AS are transmitted to device No. 9 by 101 times.

13. RED/ V

Format RED/ V_1<, V_2, . . . , V_k>

V_i: *Variable for storing data*

Function MZ-80B checks whether it has been designated as a listener and, if designated, receives data and stores them in variables.
If it is not designated as a listener. an error occurs.
The variable for storing data is the same as that of statement "RED n, V." This statement, like statement "WRT/V," is practical when receiving data from the same talker many times.



Example

```
10 DIM AS (100)
20 RED 13, AS (0)
30 FOR I = 1 TO 100
40 RED/AS (I)
50 NEXT I
```

Data from device No. 13 is received by 101 times and stored in AS( ) each time.

14. CMDW A$ *(Command Write)*

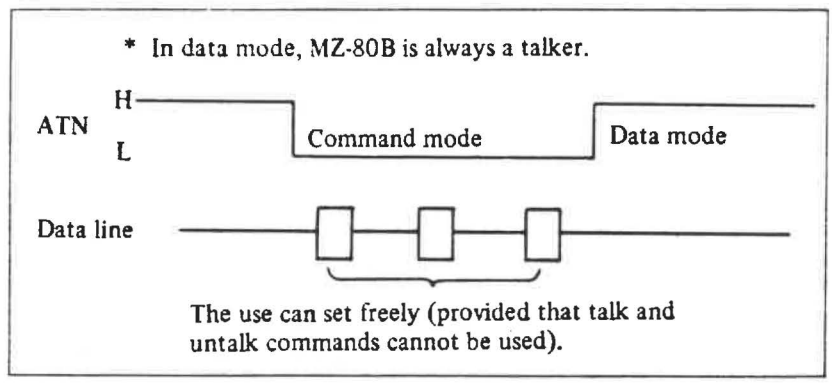
Format

CMDW *A$*

A$: transmit data

Function

MZ-80B designates itself as a talker in command mode, and sets data mode after sending the string corresponding to the command designated by *A$* to the bus.



Description

- Do NOT use UNT (5FH) or TAD (40H~ 5EH) in *A$* of this statement.
- Command mode only has to be set once, if this statement is used for repeated data transfer between the same listener and the same talker.
- Addressing instruments with secondary addresses is permitted.
- CMDW statement can be used by the user freely. However, a lack of understanding of the standards (IEC and IEEE-488) may result in total inoperation.

Example

```
10 AS=CHRS(S3F)+CHRS(S25) .... UNL code and address code to designated
                                the plotter as listener.
20 BS="LBAAA"+CHRS(S03)+CHRS(S3B) ..... Data + Delimiter
30 CMDW AS ..... Designates MZ-80B as talker and plotter as listener.
40 FOR I=0 TO 100
50 WRT/ BS ..... Writes data 101 times.
60 NEXT
```

15. CMDR B$ *(Command Read)*

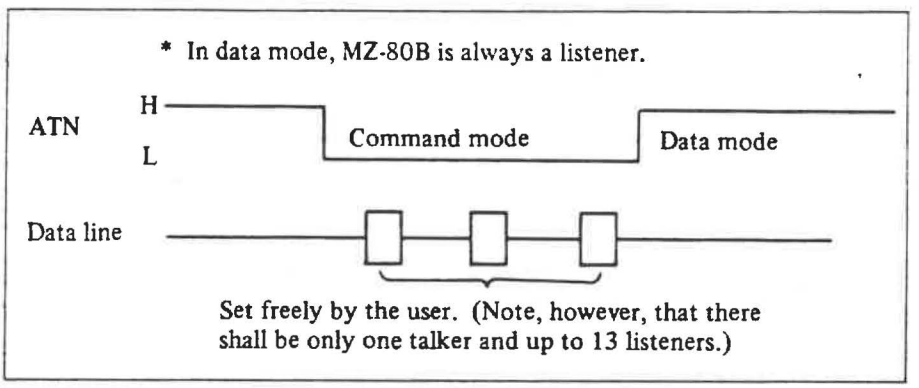
Format

CMDR *B$*

B$: receive data

Function

MZ-80B designates itself as listener in command mode, transmits the string corresponding to the command designated by *B$* to the bus, and sets data mode.



Description

- Since MZ-80B is designated as a listener, the user can designate as many as 13 listeners if one talker is designated.
- This statement can be conveniently used for command designation for repeated data transfer between the same listener and talker.
- This statement can be used for transmitting secondary address.
- CMDR statement can be used by the user freely. However a lack of understanding of the standards (IEC and IEEE-488) may result in total inoperation.

Example

```
10 AS=CHRS(S3F)+CHRS(S5F)+CHRS(S51)
                ↑          ↑          ↑
               UNL        UNT      Talk address
20 CMDR AS ..............Designates MZ-80B as listener and HP-3325A
                         (F.G) as talker.
30 RED/ BS ..............Accomodates data from talker in BS.
40 PRINT BS
```

16. ON SRQ

Format

ON SRQ *ln*

ln: line number

Function

If this statement is executed first, jump will be made to the relevant processing routine following the line number indicated by ***ln*** when an instrument on the bus makes a service request.

Description

This statement is valid only once. It must be set again, if necessary, after execution of the processing routine.

17. SPOL n, A *(Serial Polling)*

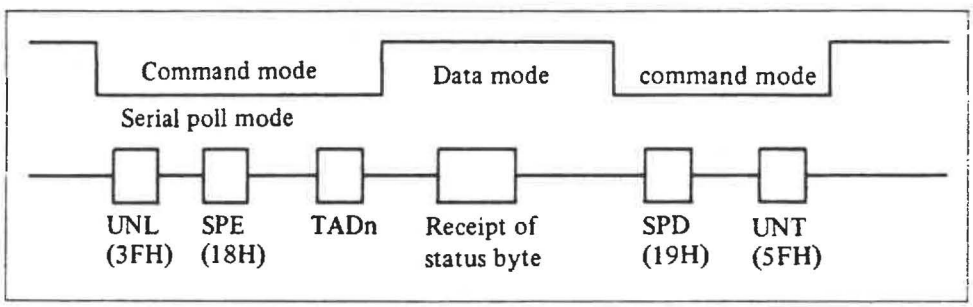
Format

SPOL *n*, *A*

n: device number (talker) *A: status byte*

Function

MZ-80B as controller clears listener, transmits SPE command followed by the talker address for instrument designated by ***n***, and sets data mode.
In data mode, MZ-80B receives status byte from the designated talker and accomodates it in variable *A*. MZ-80B sets command mode again, transmits SPD and UNT commands, and cancels serial poll mode.



Description

There shall be only one ***n***. Device indicated by ***n*** (expressed in a decimal number from 0 to 30) shall refer to the instrument with talker function.

Example

```
10 ON SRQ 150 : N=5
20 ICL : REN : DCL
30 WRT 5, "IM252, 32, 16;" ..... .Generates SRQ when an error is caused.
40 WRT 5, "SP1;SP2;SP9;" ..... .Error is generated with SP9.
50 WRT 5, "IN;" : GOTO 900 ..... .Error lamp goes out.
150 SPOL N, A ..... .Accomodates the status of talker designated by N in A.
160 IF A >= 64 THEN 50 ..... .If A is 40H or more, checks the source of request.
170 N=N+1 : GOTO 150 ..... .Checks the next talker
900 END
```

18. PPC n, l *(Parallel Poll Configure)*

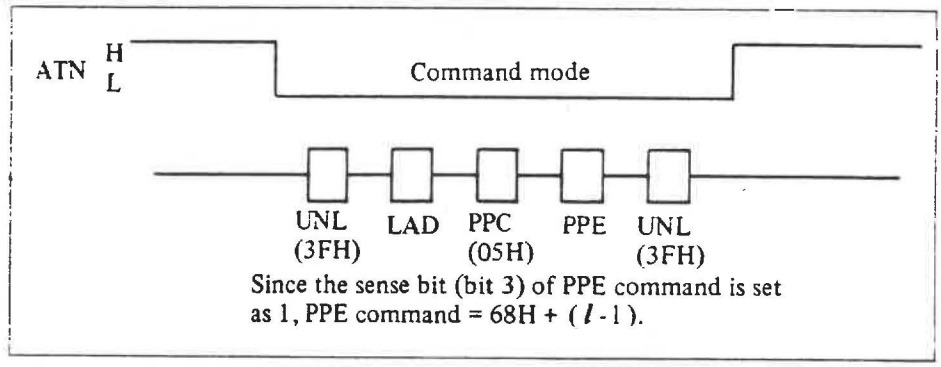
Format

PPC *n*, *l*

n: device number (listener) *l: data line No.*

Function

MZ-80B as controller sets the bus in command mode to clear listener, transmits the listener address for instrument designated by *n*, transmits PPC command, transmits PPE command expressed by (*l* -1) + 68H, and assigns data lines to which the instrument designated as listener responds in parallel poll. Finally, it clears the listener.



Description

- PPC statement must be executed before PPOL is executed in the program.
- In setting the sense bit of PPE command at 0, transmit the command using CMDW or CMDR in accordance with the timing shown above (with 60H + (*l* -1) for PPE only).
- *n* shall be a decimal number from 0 through 30.
- *l* shall be a decimal number from 1 through 8 and indicates the data line No. that corresponds.
- This statement is valid only when subset of PP function is PP1. It is unvalid if PP2 (responding bit is determined by hardware) is adopted.
- In order to cancel PPE command, PPD(70H) command must be transmitted. (Use CMDW or CMDR.)

19. PPOL A *(Parallel Poll)*

Format

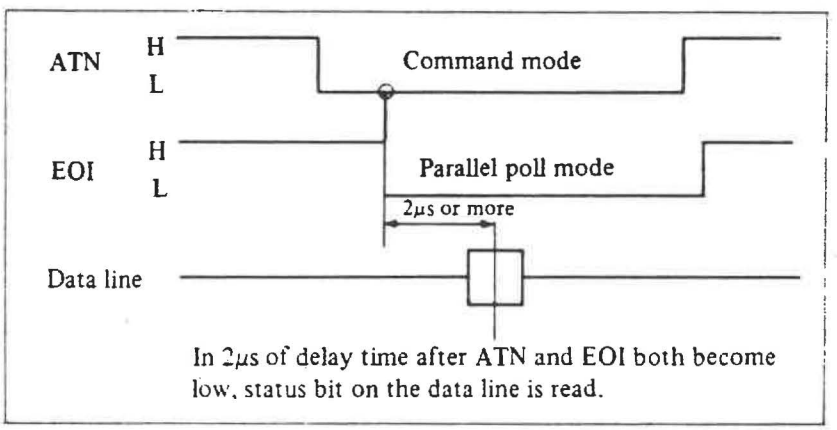
PPOL *A*

A: status bits

Function

MZ-80B as controller sets both ATN and EOI lines at low level at any point in time, and executes parallel mode.

Only on this occasion, 3-line handshake is not used. Therefore, after waiting for about 2μs until the listener sets the status bit, MZ-80B accomodates the bit pattern on the data line in variable *A*.



Description

Variable *A* is a decimal number within the range of 0 ~ 255.

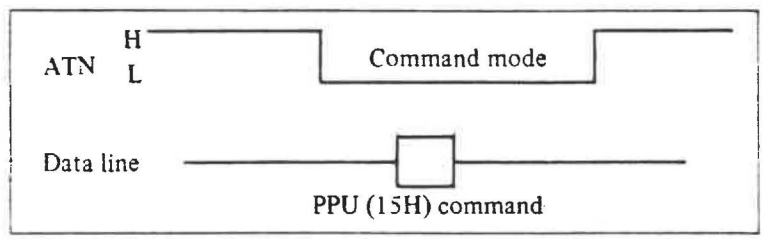
Example

```
10 ON SRQ 100
20 ICL : REN : DCL
30 WRT 5, "IM252, 32, 16;" .....Generates SRQ when an error occurs.
40 WRT 5, "SP1; SP2; SP7;" .....Since an error occured with SP7 and
                                SRQ was generated.
50 WRT 5, "SP0; IN;" : GOTO 900 ..Plotter is initialized and the execution
                                is over.
100 PPOL A ..................... Execution of parallel poll.
110 IF A=4 THEN 50 ..............OK if A=4.
120 PRINT "ERROR" ..............Error indication if A≠4.
900 END
```

20. PPU *(Parallel Poll Unconfigure)*

Format PPU

Function MZ-80B issues PPU command in command mode to cancel parallel poll mode.



Description In order to execute parallel poll again, the statement "PPC n, 1" must be executed.

21. GPIBM n

Format GPIBM *n*

n: device number

Function MZ-80B is set as system controller and active controller, and designates the decimal number (0 ~ 30) indicated by *n* as its own address.

22. EOIW n

Format EOIW n_1 or EOIW n_1, n_2

n_1, n_2: *delimiter*

Function Can be designated by one or 2 decimal number(s) as delimiter for data transmission.

Description

- EOI becomes low only for a short time before and after transmission of the last delimiter.
- Decimal numbers in excess of two will be ignored.

23. EOIR n

Format EOIR n_1 or EOIR n_1, n_2

n_1, n_2: *delimiter*

Function Can be designated by one or 2 decimal number(s) as delimiter for data receiving.

Description

- If EOI line is at low level, operation will be finished after the data residing on the data line at that time are received.
- Decimal numbers in excess of 2 shall be ignored.

Cautions for Programming

1. Address

Each measuring instrument or device has its own address, which can be altered. Caution must be taken since existence of instruments with the same address within one system may cause wrong operation.

2. Data delimiter

Each instrument has a code (or delimiter) to judge whether data have ended. There are a variety of delimiters. In an extreme case, like instruments may have different delimiters, depending on operation modes.
In MZ-80B, two bytes 0DH–0AH are defined as delimiters in the initial setting. However, a compliance with the delimiter of the instrument used must be achieved. Otherwise, data transfer will be stopped midway.
Delimiters are specified in the Operation Manual of measuring instruments, so refer to such manuals.

3. Sequence of statement execution

ON SRQ in serial poll, "PPC n, l" in parallel poll, address setting, designation of delimiter, ICL, REN, etc. must be executed at the beginning of programs. Otherwise, they will have no meaning.

4. When instruments do not respond

Stop the program using the BREAK key, rectify the program error, and execute (RUN) the program again.

5. WAIT routine

Wait time until the instrument gets ready for data transmit/receive must be set in the instrument. Since MZ-80B has no WAIT statement, set desired wait time with the use of FOR-NEXT loop.

6. When the opposite side has no PCT function

If the controller, which received the right to control from MZ-80B with the use of PCT statement, does not have the function to return the right to MZ-80B, execute ICL. In this way, the opposite controller will be initialized, and the right to control will be returned to MZ-80B.
However, MZ-80B must be a system controller.

Chapter 2
RS-232C Statements

This chapter contains information on the BASIC statements that are used to control peripheral equipment with the serial interface card MZ-8B103.

Reference the Serial Interface Manual.

1. RSMODE

Format RSMODE *a*, R*b*, T*c*, M*d*. RX*e*

a: Channel Specification

a	Channel
A	A Channel
B	B Channel

b: Specification of the number of bits for the received character.

b	bits/character
5	5
6	6
7	7
8	8

c: Specification of the number of bits for the transmitted character.

c	bits/character
5	5
6	6
7	7
8	8

d: Parity bit specification and number of stop bits specification

d	parity	stop bits
69	odd	1
70	none	
71	even	
73	odd	1½
74	none	
75	even	
77	odd	2
78	none	
79	even	

e: Receive active/inactive specification

e	Receive
0	Inactive
1	Active

Function Setting of the various modes is performed using the above parameters.

Description

- Although it is not necessary to specify all of the parameters, specification of the receive active/inactive status must be made after specification of the various parameters.
- Specification of mode setting parameters other than those given in *a* through *e* are not allowed.
- When BASIC is initialized, all channels are set for the following modes.

b, *c*	8	8 bits per character
d	79	Even parity/Stop bits 2
e	0	Receive inactive

Example 10 RSMODE A, RX1 .....Enables reception in channel A.

2. RSO

Format RSO *x A$*

x: Channel specification (A or B).

A$: Specify transmission data using a string variable

Function Transmit the data specified using *A$* to channel *x*

Example

```
10 XS = "Demonstration"
20 RSO B XS ..... .Transmit the data in XS to channel B.
```

3. RSI

Format RSI *x A$*

x: Channel specification (A or B)

A$: A string variable which contains the received data

Function Data is received from channel *x* and stored in the string variable *A$*.

Example

```
10 RSMODE A, RX1 ..... .Activates channel A for reception.
20 RSI A BS ..... .Receives data from channel A.
30 PRINT BS
```

APPENDIX

The Appendix includes the following;

- *ASCII code Table . . . . . Table A. 1*
- *DISK BASIC interpreter SB-6511 Error Message Table . . . . . Table A. 2*
  This table lists all the possible errors which may occur during program execution. The interpreter notifies the operator when an error occurs during program execution or operation in the direct mode, by outputting the corresponding error number.
- *Memory Map*
- *Handling diskettes*

A.1 ASCII Code Table

Decimal	Hexa-decimal	Charac-ter	Decimal	Hexa-decimal	Charac-ter	Decimal	Hexa-decimal	Charac-ter	Decimal	Hexa-decimal	Charac-ter	Decimal	Hexa-decimal	Charac-ter
0	00	NULL	26	1A		52	34	4	78	4E	N	104	68	h
1	01	↓	27	1B		53	35	5	79	4F	O	105	69	i
2	02	↑	28	1C		54	36	6	80	50	P	106	6A	j
3	03	→	29	1D		55	37	7	81	51	Q	107	6B	k
4	04	←	30	1E		56	38	8	82	52	R	108	6C	l
5	05	HOME	31	1F	▒	57	39	9	83	53	S	109	6D	m
6	06	CLR	32	20		58	3A	:	84	54	T	110	6E	n
7	07	DEL	33	21	!	59	3B	;	85	55	U	111	6F	o
8	08	INST	34	22	"	60	3C	<	86	56	V	112	70	p
9	09	GRPH	35	23	#	61	3D	=	87	57	W	113	71	q
10	0A	SFT LOCK	36	24	$	62	3E	>	88	58	X	114	72	r
11	0B		37	25	%	63	3F	?	89	59	Y	115	73	s
12	0C	RVS	38	26	&	64	40	@	90	5A	Z	116	74	t
13	0D		39	27	'	65	41	A	91	5B	[	117	75	u
14	0E	SCRIPT	40	28	(	66	42	B	92	5C	\	118	76	v
15	0F	RVS CANCEL	41	29	)	67	43	C	93	5D	]	119	77	w
16	10		42	2A	*	68	44	D	94	5E	^	120	78	x
17	11		43	2B	+	69	45	E	95	5F	¯	121	79	y
18	12		44	2C	,	70	46	F	96	60	`	122	7A	z
19	13		45	2D	—	71	47	G	97	61	a	123	7B	{
20	14		46	2E	.	72	48	H	98	62	b	124	7C	|
21	15		47	2F	/	73	49	I	99	63	c	125	7D	}
22	16		48	30	0	74	4A	J	100	64	d	126	7E	~
23	17		49	31	1	75	4B	K	101	65	e	127	7F	↵
24	18		50	32	2	76	4C	L	102	66	f			
25	19		51	33	3	77	4D	M	103	67	g			

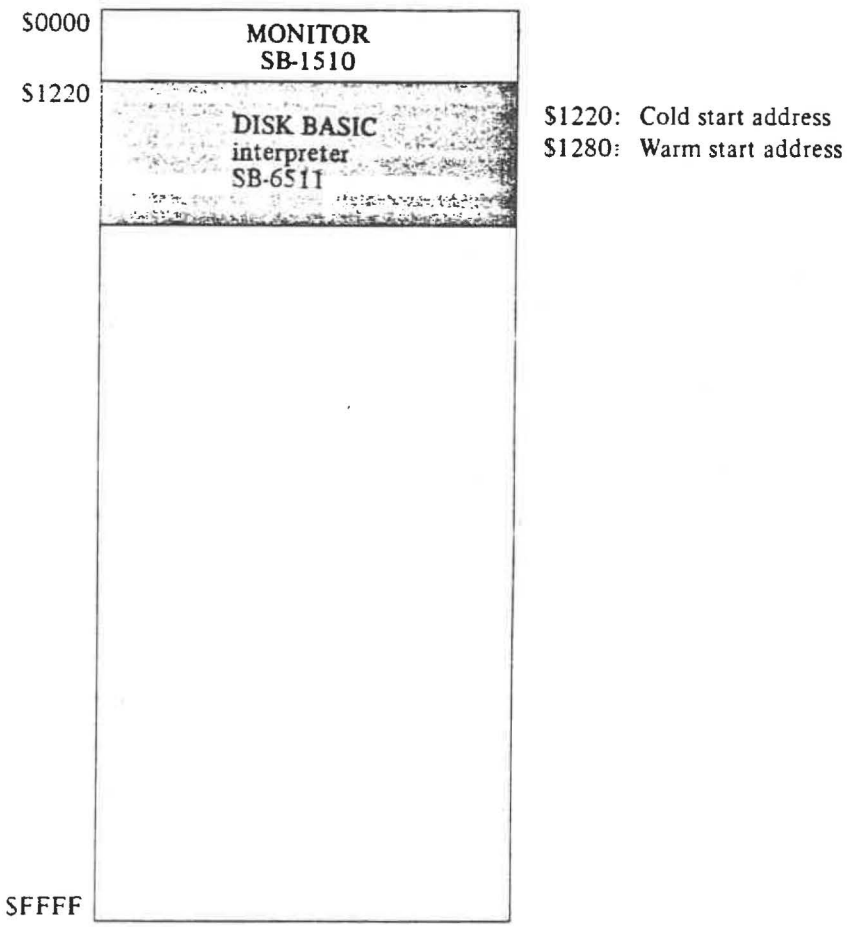
Decimal	Hexa-decimal	Charac-ter	Decimal	Hexa-decimal	Charac-ter	Decimal	Hexa-decimal	Charac-ter	Decimal	Hexa-decimal	Charac-ter	Decimal	Hexa-decimal	Charac-ter
128	80		154	9A		180	B4		206	CE		232	E8	
129	81		155	9B		181	B5		207	CF		233	E9	
130	82		156	9C		182	B6		208	D0		234	EA	
131	83		157	9D		183	B7		209	D1		235	EB	
132	84		158	9E		184	B8		210	D2		236	EC	
133	85		159	9F		185	B9		211	D3		237	ED	
134	86		160	A0		186	BA		212	D4		238	EE	
135	87		161	A1		187	BB		213	D5		239	EF	
136	88		162	A2		188	BC		214	D6		240	F0	
137	89		163	A3		189	BD		215	D7		241	F1	
138	8A		164	A4		190	BE		216	D8		242	F2	
139	8B		165	A5		191	BF		217	D9		243	F3	
140	8C		166	A6		192	C0		218	DA		244	F4	
141	8D		167	A7		193	C1		219	DB		245	F5	
142	8E		168	A8		194	C2		220	DC		246	F6	
143	8F		169	A9		195	C3		221	DD		247	F7	
144	90		170	AA		196	C4		222	DE		248	F8	
145	91		171	AB		197	C5		223	DF		249	F9	
146	92		172	AC		198	C6		224	E0		250	FA	
147	93		173	AD		199	C7		225	E1		251	FB	
148	94		174	AE		200	C8		226	E2		252	FC	
149	95		175	AF		201	C9		227	E3		253	FD	
150	96		176	B0		202	CA		228	E4		254	FE	
151	97		177	B1		203	CB		229	E5		255	FF	
152	98		178	B2		204	CC		230	E6				
153	99		179	B3		205	CD		231	E7				

A.2 Error Message Table

	Error No.	Meaning
General	1	Syntax error
	2	Operation result overflow
	3	Illegal data
	4	Data type mismatch
	5	String length exceeded 255 characters
	6	Insufficient memory capacity
	7	The size of a defined array exceeded its previous definition.
	8	The length of a BASIC text line was too long.
	9	
	10	The number of GOSUB nesting levels exceeded 15.
	11	The number of FOR-NEXT nesting levels exceeded 15.
	12	The number of function levels exceeded 6.
	13	NEXT was used without a corresponding FOR.
	14	RETURN was used without a corresponding GOSUB.
	15	An undefined function was used.
	16	An unused reference line number was specified in a statement.
	17	The CONT command cannot be executed.
	18	A write statement was issued to the BASIC control area.
	19	Direct mode commands and statements are mixed together.
	20	The RESUME statement cannot be executed.
	21	A RESUME statement was used without a corresponding error process.
	22	
	23	
	24	A READ statement was used without a corresponding DATA statement.
	25	The number of SWAP levels exceeded 1.
	26	
	27	
	28	
RS-232C	29	Framing error
	30	Overrun error
	31	Parity error
	32	Data transmission is impossible. (The transmit buffer is not empty.)
	33	Buffer overflow
GP-IB	34	A system controller command was used when there was no system controller.
	35	Card or cable is not connected.

	Error No.	Meaning
GP-IB	36	3-line handshake NDAC and NRFD are both High level.
	37	Address code setting error
	38	Both talker and listener are active on the card.
	39	Listener or talker is not active.
Disk	40	File was not found.
	41	Disk drive hardware error.
	42	A file name already in use was defined again.
	43	OPEN, DELETE, or RENAME statements were issued to an open file.
	44	An unopened file was referenced or a CLOSE or KILL statement was issued to it.
	45	
	46	A protected file was accessed for writing.
	47	
	48	
	49	
	50	The disk drive is not ready.
	51	The total number of files on a volume exceeded 63.
	52	Volume number error
	53	File space on the diskette is insufficient.
	54	A diskette which has not been initialized was loaded.
	55	The number of data of a BSD file exceeded 64K bytes.
	56	Data error occurred on an FDC routine call.
	57	The diskette cannot be used.
Cassette or Disk	58	
	59	
	60	Illegal file name was specified.
	61	Illegal file mode was specified.
	62	
	63	Out of file
	64	Illegal logical number was specified.
Printer	65	The printer is not ready.
	66	Printer hardware error
	67	Out of paper
	68	
	69	
	70	Check sum error

A.3 Memory Map



A.4 Handling diskettes

The master diskette must be handled especially carefully. Make a submaster diskette by means of the diskette-copy program in the OBJ file "Utility" on the master diskette. Be sure to keep the master diskette in a safe place.

All optional blank diskettes supplied by the Sharp Co. are not initialized. Be sure to initialize them before use.

Notes on handling of diskettes

- Fingerprints on a diskette may render it permanently unusable. Never touch the diskette surface through the head window.
- Insert the diskette straight into the drive until it stops, then close the front door gently. Rough handling may damage the diskette.
- Do not fold or bend the diskette since this may render it unusable.
- Write the index label before it is affixed to the jacket. If it is written after it is affixed to the jacket, use a felt marker or other soft tip pen.
- Ashes and drinks are the most common contaminants to guard against.
- Ambient temperature: 4 ~ 53°C

  When the ambient temperature is more than 53°C the jacket may be deformed. Do not place the diskette in a place where it is exposed to direct sun light or where the temperature may exceed 53°C.

  When using the diskette, ensure that the temperature range described on the pritective envelope is observed. Environmental conditions may differ between the places of storage and operation. This requires the diskette to be placed in the proper operating environment for a while before use.

Notes on storing diskettes

- Keep the diskettes away from magnets. Even a magnetic ring or magnetic necklace may damage data on the diskette. Electrical equipment such as the display unit of the computer, a cassette tape recorder, or a TV set generates a magnetic field, so keep diskettes away from such euipment.
- Keep the diskette in the envelope supplied. Make it a habit to put the diskette in the envelope immediately after it has been taken out of the drive. This will prevent almost all problems which result from the careless handling of diskettes. The master diskette must be handled especially carefully.

  The envelopes supplied are made of special materials and help guard against static electricity and moisture.
- When storing diskettes for a long time, keep the envelopes in the storage case. Be sure the envelopes are stored vertically in the storage case. Do not incline or bend the envelope. The master diskette is not supplied with a storage case.
- Do not attach paper clips or the like to diskettes.
- Do not place any heavy objects on diskettes.